UN1101.002: Calculus I, Midterm 1 Make up extra work: Spring 26

NAME: UNI:

This is due on Saturday Feb 28 at 11:59pm. Late work will not be accepted. Write your answers LEGIBLY on this paper, and submit on Courseworks. Do this ONLY if you got < 12 on Midterm I.

Ex 1 (2 points each) Find the limits (finite or infinite) if they exist. Show some intermediate steps in your calculations, and name the important rules that you are using. (One point for a correct answer, and 1 point for good explanation.)

(i) $\lim_{x \to 1} \frac{x^2+3x-4}{x-x^3} =$

(ii) $\lim_{x \to 1^+} \frac{x^2+3}{x-x^3} =$

(iii) $\lim_{x \to 1/e} \frac{\ln(\frac{1}{x})}{x} =$

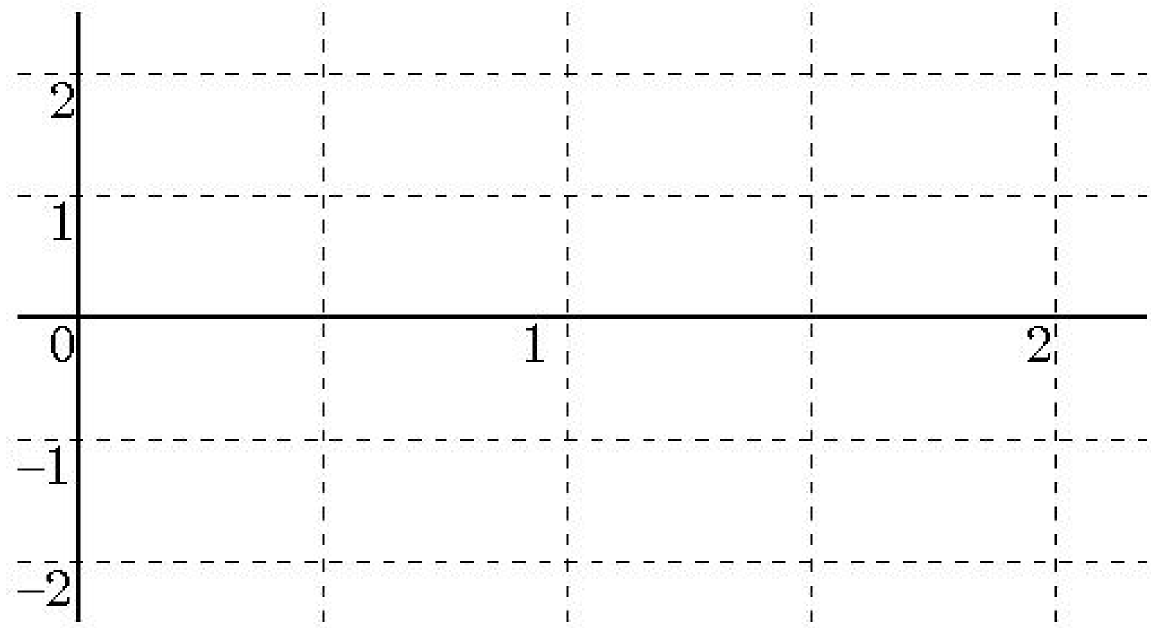
2. (4 points) Let $f(x) = 2 - \sin x$ have domain $[-\pi/2, \pi/2]$.
(i) What is the range of f?

(ii) Find a formula for its inverse f^{-1} and evaluate $f^{-1}(3/2)$.

3. (4 points) (i) Find all solutions of the equation $ex - e^{3\ln(x)-2} = 0$.

(ii) Why is $x = 0$ not a solution?

4. (3 points) (i) Draw a careful graph of the functions $f(x) := \sin(\pi x)$ and $g(x) := x\sin(\pi x)$ on the grid below, labelling their values at $x = 0, 1/2, 1, 3/2, 2$.



5. (i) (1 point) State the intermediate Value theorem.

(ii) (2 points) Use it to show that there is a point x in $[0, \pi]$ such that $\sin(2x) = 2\cos(x)$.